CHAPTER # 2 :-

Basic Structures:

SETS, FUNCTIONS,

SEQUENCES, SUMS, MATRICES

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Exercise 2.1

1. List the members of these sets.

(a) $\{x \mid x$ is a real number such that $x^2 = 1\}$

$\{-1, -1\}$

(b) $\{x \mid x$ is a positive integer less than $12\}$

$\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$

(c) $\{x \mid x$ is the square of an in integer and $x < 100\}$

$\{0, 1, 4, 9, 16, 25, 36, 49, 64, 81\}$

(d) $\{x \mid x$ is an integer such thant $x^2 = 2\}$

ϕ

2. Use set builder notation to give a description of each of these sets.

(a) $\{0, 3, 6, 9, 12\}$

$\{x \in N \mid x$ is a multiple of and $x \leq 12\}$

(b) $\{-3, -2, -1, 0, 1, 2, 3\}$

$\{x \in Z \mid -3 \leq x \leq 3\}$

(c) $\{m, n, o, p\}$

$\{x \mid x$ is a letter in the alphabet from m to p$\}$

3. For each of these pairs of sets, determine whether the first is the subset of the second, the second is the subset of the first, or niether is the subset of the other.

(a) the set of airline flights from New York to New Delhi, the set of nonstop airlines flights from New York to New Delhi

The first is the subset of the second, but second is not.

(b) the set of people who speak English. the set of people who speak Chinese

Niether is the subset of the other.

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Two sets are **equal** if and only if they have the same elements. Therefore, if A and B are sets then A and B are equal if and only if $\forall x(x \in A \leftrightarrow x \in B)$. We write $A = B$ if A and B are equal sets.

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e (c) the set of the flying squirrels, the set of living creatures that can fly

The first is the subset of the second but the second is not the subset of the first.

4. For each of these pairs of sets, determine whether the first is the subset of the second, the second is the subset of the first, or niether is the subset of the other.

(a) the set of people who speak English, the set of people who speak English with an Australian accent

The second is the subset of the first but the first is not the subset of the second.

(b) the set of fruits, the set of citrus fruits

The second is the subset of the first but the first is not the subset of the second.

(c) the set of students studying discrete mathematics, the set of students studying data structures.

Niether is the subset of the other.

5. Determine whether each of these pairs of sets are equal.

(a) $\{1,3,3,3,5,5,5,5,5\}$, $\{5,3,1\}$

The sets are equal.

(b) $\{\{1\}\}$, $\{1,\{1\}\}$ The first set contain 1, while the second set contain 2 elements.

The sets are not equal.

(c) ϕ , $\{\phi\}$

The sets are not equal.

6. Suppose that $A=\{2,4,6\}$, $B=\{2,6\}$, $C=\{4,6\}$ and $D=\{4,6,8\}$. Determine which of these sets are

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If a set A is a subset of a set B but $A \neq B$, we write $A \subset B$ and say that A is a **proper subset** of B. For $A \subset B$ to be true, it must be the case that $A \subseteq B$ and there must exist an element x of B which is not an element of A.

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subsets of which other of these sets.
$A \subseteq A, B \subseteq A, B \subseteq B, C \subseteq A, C \subseteq C, C \subseteq D,$
$D \subseteq D$

7. For each of the following sets, determine whether 2 is an element of that set.
(a) $\{x \in R \mid x$ is an integer greater than $1\}$
Yes
(b) $\{x \in R \mid x$ is the square of an integer$\}$
No
(c) $\{2, \{2\}\}$ Yes
(d) $\{\{2\}, \{\{2\}\}\}$ No
(e) $\{\{2\}, \{2, \{2\}\}\}$ No
(f) $\{\{\{2\}\}\}$ No

8. For each of these sets in Exercises 7, determine whether $\{2\}$ is an element of that set.
(a) No
(b) No
(c) Yes
(d) Yes
(e) Yes
(f) No

9. Determine whether each of these statements is true or false.
(a) $0 \in \phi$ False
(b) $\phi \in \{0\}$ False
(c) $\{0\} \subset \phi$ False
(d) $\phi \subset \{0\}$ True
(e) $\{0\} \in \{0\}$ False
(f) $\{0\} \subset \{0\}$ False
~~(g) $\{\{\phi\}\} \subset \{\{\phi\}, \{x$~~
For a set to be a proper subset of another, they can't be equal

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(g) $\{\phi\} \subseteq \{\phi\}$ True

10. Determine whether these statements are true or false.

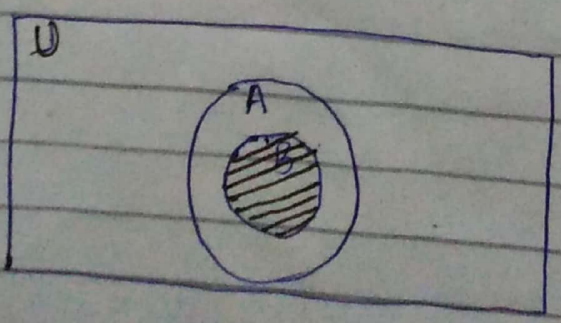
(a) $\phi \in \{\phi\}$ True
(b) $\phi \in \{\phi, \{\phi\}\}$ True
(c) $\{\phi\} \in \{\phi\}$ False
(d) $\{\phi\} \in \{\{\phi\}\}$ True
(e) $\{\phi\} \subset \{\phi, \{\phi\}\}$ True
(f) $\{\{\phi\}\} \subset \{\phi, \{\phi\}\}$ True
(g) $\{\{\phi\}\} \subset \{\{\phi\}, \{\phi\}\}$ False

11. Determine whether each of these statements is true or false.

(a) $\{x \in \{x\}$ True
(b) $\{x\} \subseteq \{x\}$ True
(c) $\{x\} \in \{x\}$ False
(d) $\{x\} \in \{\{x\}\}$ True
(e) $\{\phi \subseteq \{x\}$ True
(f) $\phi \in \{x\}$ False

12. Use a Venn diagram to illustrate the subset of odd integers in the set of all positive integers not exceeding 10.

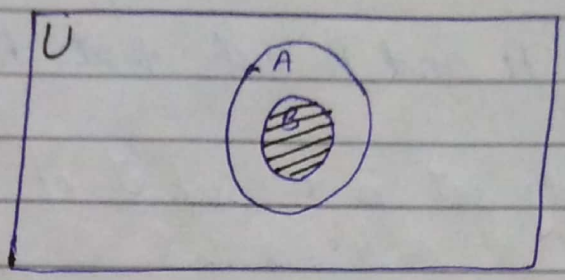
Let $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
and $B = \{1, 3, 5, 7, 9\}$



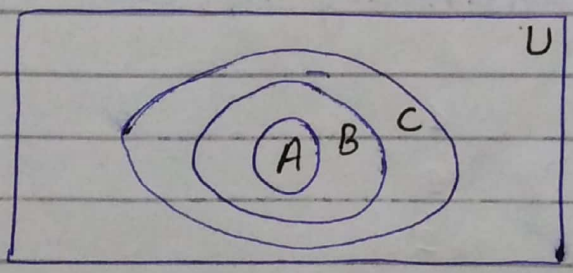
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13. Use a Venn diagram to illustrate the set of all months of the year whose names do not contain the letter R in the set of all months of the year.

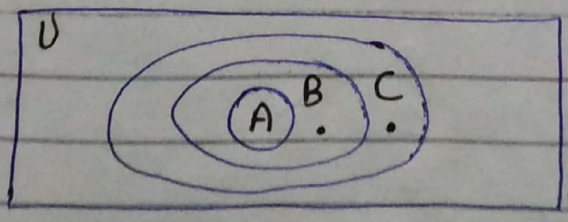
Let A = {January, February, March, ..., December}
and B = {May, June, July, August}



14. Use a Venn diagram to illustrate the relationship $A \subseteq B$ and $B \subseteq C$.

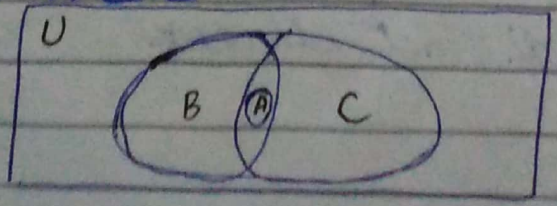


15. Use a Venn diagram to illustrate the relationship $A \subset B$ and $B \subset C$.



Dot shows that there is atleast an element in C which is not in B and in B which is not in A

16. Use a Venn diagram to illustrate the relationship $A \subset B$ and $A \subset C$



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17. Suppose that A, B and C are sets such that $A \subseteq B$ and $B \subseteq C$. Show that $A \subseteq C$.

Suppose that $x \in A$. Because $A \subseteq B$, this implies that $x \in B$. Because $B \subseteq C$, we see that $x \in C$. Because $x \in A$ implies that $x \in C$, it follows that $A \subseteq C$.

18. Find two sets A and B such that $A \in B$ and $A \subseteq B$.

Since the empty set is a subset of every set, we just need to take a set B that contains ϕ as an element. Thus we can let $A = \phi$ and $B = \{\phi\}$ as the simple example.

19. What is the cardinality of each of these sets?

(a) $\{a\}$ 1

(b) $\{\{a\}\}$ 1

(c) $\{a, \{a\}\}$ 2

(d) $\{a, \{a\}, \{a, \{a\}\}\}$ 3

20. What is the cardinality of each of these sets?

(a) $\emptyset$ 0

(b) $\{\emptyset\}$ 1

(c) $\{\emptyset, \{\emptyset\}\}$ 2

(d) $\{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$ 3

21. Find the power set of each of these sets, where a and b are distinct elements.

(a) $\{a\}$

$\{\phi, \{a\}\}$

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(b) $\{a, b\}$

$\{\emptyset, \{a\}, \{b\}, \{a, b\}\}$

(c) $\{\emptyset, \{\emptyset\}\}$

$\{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\emptyset\}\}\}$

23. How many elements does each of these sets have where a and b are distinct elements?

(a) $P(\{a, b, \{a, b\}\})$ 8

(b) $P(\{\emptyset, a, \{a\}, \{\{a\}\}\})$ 16

(c) $P(P(\emptyset))$ 2

24. Determine whether each of these sets is the power set of a set, where a and b are disbinct elements.

(a) $\emptyset$

The power set of all the sets includes at least the empty set, so the power set cannot be empty. Thus $\emptyset$ is not the power set of any set.

(b) $\{\emptyset, \{a\}\}$

This is the power set of $\{a\}$

(c) $\{\emptyset, \{a\}, \{\emptyset, a\}\}$

This set has 3 elements. Since 3 is not a power of 2, this set cannot be the power set of anyset.

(d) $\{\emptyset, \{a\}, \{b\}, \{a, b\}\}$

This is the power set of $\{a, b\}$.

25. Prove that $P(A) \subseteq P(B)$ if and only if $A \subseteq B$.

Suppose $a \in A$, Then $\{a\} \subseteq A$, so $\{a\} \in P(A)$. Since $P(A) \subseteq P(B)$, it follows that $\{a\} \in P(B)$, which means that $\{a\} \subseteq B$. But this implies $a \in B$, as desired.

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27. Let $A=\{a,b,c,d\}$ and $B=\{y,z\}$. Find

(a) $A \times B$

$A \times B = \{(a,y),(b,y),(c,y),(d,y),(a,z),(b,z),(c,z),(d,z)\}$

(b) $B \times A$

$B \times A = \{(y,a),(y,b),(y,c),(y,d),(z,a),(z,b),(z,c),(z,d)\}$

30. Suppose that $A \times B = \phi$, where A and B are sets. What can you conclude?

We can conclude that either $A=\phi$ or $B=\phi$ or both are empty sets.

3

31. Let A be a set. Show that $\phi \times A = A \times \phi = \phi$

$\phi \times A = \{(x,y) \mid x \in \phi \text{ and } y \in A\} = \phi = \{(x,y) \mid x \in A \text{ and } y \in \phi\} = A \times \phi$

32. Let $A=\{a,b,c\}$, $B=\{x,y\}$ and $C=\{0,1\}$. Find

(a) $A \times B \times C$

$A \times B \times C = \{(a,x,0),(a,x,1),(a,y,0),(a,y,1),(b,x,0),(b,x,1),(b,y,0),(b,y,1),(c,x,0),(c,x,1),(c,y,0),(c,y,1)\}$

(b) $C \times B \times A$

$C \times B \times A = \{(0,x,a),(0,x,b),(0,x,c),(0,y,a),(0,y,b),(0,y,c),(1,x,a),(1,x,b),(1,x,c),(1,y,a),(1,y,b),(1,y,c)\}$

(c) $C \times A \times B$

$C \times A \times B = \{(0,a,x),(0,a,y),(0,b,x),(0,b,y),(0,c,x),(0,c,y),(1,a,x),(1,a,y),(1,b,x),(1,b,y),(1,c,x),(1,c,y)\}$

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(d) $B \times B \times B$

$B \times B \times B = \{(x,x,x), (x,x,y), (x,y,x), (x,y,y), (y,x,x), (y,x,y), (y,y,x), (y,y,y)\}$

33. Find A^2 if

(a) $A = \{0,1,3\}$

$A^2 = A \times A = \{(0,0), (0,1), (0,3), (1,0), (1,1), (1,3), (3,0), (3,1), (3,3)\}$

(b) $A = \{1,2,a,b\}$

$A^2 = A \times A = \{(1,1), (1,2), (1,a), (1,b), (2,1), (2,2), (2,a), (2,b), (a,1), (a,2), (a,a), (a,b), (b,1), (b,2), (b,a), (b,b)\}$

34. Find A^3 if

(a) $A = \{a\}$

$A^3 = A \times A \times A = \{(a,a,a)\}$

(b) $A = \{0,a\}$

$A^3 = A \times A \times A = \{(0,0,0), (0,0,a), (0,a,0), (0,a,a), (a,0,0), (a,0,a), (a,a,0), (a,a,a)\}$

35. How many different elements does $A \times B$ have if A has m elements and B has n elements?

mn elements

36. How many different elements does $A \times B \times C$ have if A has m elements, B has n elements, and C has p elements?

mnp elements.

37. How many different elements A^n have when A has m elements and n is positive integer?

m^n

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38. Show that $A \times B \neq B \times A$ when A and B are non-empty, unless $A = B$.

Suppose that $A \neq B$ and neither A nor B is empty. We must prove that $A \times B \neq B \times A$. Since $A \neq B$, either we can find an element $x \in A$ that is in A, but not B, or vice versa. The two cases are similar, so without loss of generality, let us assume that x is in A but not B. Also, since B is not empty, there is some element $y \in B$. Then (x, y) is in $A \times B$ by definition but it is not in $B \times A$ since $x \notin B$. Therefore $A \times B \neq B \times A$.

39. Explain why $A \times B \times C$ and $(A \times B) \times C$ are not the same.

The elements of $A \times B \times C$ consists of 3-tuples (a, b, c) where $a \in A$, $b \in B$, and $c \in C$, whereas the elements the elements of $(A \times B) \times C$ look like $((a, b), c)$ — ordered pairs, the first coordinate of which is again an ordered pair.

40. Explain why $(A \times B) \times (C \times D)$ and $A \times (B \times C) \times D$ are not the same.

The only difference between both of them is paranthesis, for both of them order pairs are differently written. i.e., for $(A \times B) \times (C \times B)$ ordered pairs are written as $((a, b), (c, d))$ where $a \in A$, $b \in B$, $c \in C$, $d \in D$ and for $A \times (B \times C) \times D$ ordered pairs are written as $(a, (b, c), d)$.

41. Translate each of these quantifications into English and determine its truth value.

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(a) $\forall x \in R(x^2 \neq -1)$

The square of a real number is never -1. True

(b) $\exists x \in Z(x^2 = 2)$

There exists an integer whose square is 2. False.

(c) $\forall x \in Z(x^2 > 0)$

The square of every integer is positive. False

(d) $\exists x \in R(x^2 = x)$

There is a real number equal to its own square. True.

42. Translate each of these quantification into English and determine its truth value.

(a) $\exists x \in R(x^3 = -1)$

There is a real number whose cube is -1. True.

(b) $\exists x \in Z(x+1 > x)$

There is an integer such that the number obtained by adding 1 to it is greater than the integer. True

(c) $\forall x \in Z(x-1 \in Z)$

For every integer, the number obtained by subtracting 1 is again an integer. True.

(d) $\forall x \in Z(x^2 \in Z)$

The square of every integer is integer. True.

43. Find the truth set of each of these predicates where domain is the set of integers.

(a) $P(x): x^2 < 3$

$\{-1, 0, 1\}$

(b) $Q(x): x^2 > x$

$Z - \{0, 1\}$

(c) $R(x): 2x + 1 = 0$

ϕ

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44. Find the truth set of each of these predicates where the domain is the set of integers.

(a) $P(x): x^3 \geq 1$

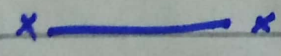
$\{1, 2, 3, \ldots\}$

(b) $Q(x): x^2 = 2$

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(c) $R(x): x < x^2$

$\{\ldots, -3, -2, -1, 2, 3, \ldots\}$



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EXCERCISE 2.2

3. Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{0, 3, 6\}$. Find

(a) $A \cup B$

$A \cup B = \{1, 2, 3, 4, 5\} \cup \{0, 3, 6\}$

$A \cup B = \{0, 1, 2, 3, 4, 5, 6\}$

(b) $A \cap B$

$A \cap B = \{1, 2, 3, 4, 5\} \cap \{0, 3, 6\}$

$A \cap B = \{3\}$

(c) $A - B$

$A - B = \{1, 2, 3, 4, 5\} - \{0, 3, 6\}$

$A - B = \{1, 2, 4, 5\}$

(d) $B - A$

$B - A = \{0, 3, 6\} - \{1, 2, 3, 4, 5\}$

$B - A = \{0, 6\}$

4. Let $A = \{a, b, c, d, e\}$ and $B = \{a, b, c, d, e, f, g, h\}$

(a) $A \cup B$

$A \cup B = \{a, b, c, d, e\} \cup \{a, b, c, d, e, f, g, h\}$

$A \cup B = \{a, b, c, d, e, f, g, h\}$

(b) $A \cap B$

$A \cap B = \{a, b, c, d, e\} \cap \{a, b, c, d, e, f, g, h\}$

$A \cap B = \{a, b, c, d, e\}$

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(c) $A - B$

$A - B = \{a, b, c, d, e\} - \{a, b, c, d, e, f, g, h\}$

$A - B = \emptyset$

(d) $B - A$

$B - A = \{a, b, c, d, e, f, g, h\} - \{a, b, c, d, e\}$

$= \{f, g, h\}$

14. **Find the sets A and B if $A - B = \{1, 5, 7, 8\}$, $B - A = \{2, 10\}$ and $A \cap B = \{3, 6, 9\}$**

$A = (A - B) \cup (A \cap B)$

$A = \{1, 5, 7, 8\} \cup \{3, 6, 9\}$

$A = \{1, 3, 5, 6, 7, 8, 9\}$

$B = (B - A) \cup (A \cap B)$

$B = \{2, 10\} \cup \{3, 6, 9\}$

$B = \{2, 3, 6, 9, 10\}$

25. **Let $A = \{0, 2, 4, 6, 8, 10\}$, $B = \{0, 1, 2, 3, 4, 5, 6\}$, and $C = \{4, 5, 6, 7, 8, 9, 10\}$. Find**

(a) $A \cap B \cap C$

$A \cap B \cap C = \{0, 2, 4, 6, 8, 10\} \cap \{0, 1, 2, 3, 4, 5, 6\} \cap \{4, 5, 6, 7, 8, 9, 10\}$

$A \cap B \cap C = \{4, 6\}$

b) $A \cup B \cup C$

$A \cup B \cup C = \{0, 2, 4, 6, 8, 10\} \cup \{0, 1, 2, 3, 4, 5, 6\} \cup \{4, 5, 6, 7, 8, 9, 10\}$

$= \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

(c) $(A \cup B) \cap C$

$(A \cup B) \cap C = (\{0, 2, 4, 6, 8, 10\} \cup \{0, 1, 2, 3, 4, 5, 6\}) \cap \{4, 5, 6, 7, 8, 9, 10\}$

$= \{0, 1, 2, 3, 4, 5, 6, 8, 10\} \cap \{4, 5, 6, 7, 8, 9, 10\}$

$= \{4, 5, 6, 8, 10\}$

(d) $(A \cap B) \cup C$

$(A \cap B) \cup C = (\{0, 2, 4, 6, 8, 10\} \cap \{0, 1, 2, 3, 4, 5, 6\}) \cup \{4, 5, 6, 7, 8, 9, 10\}$

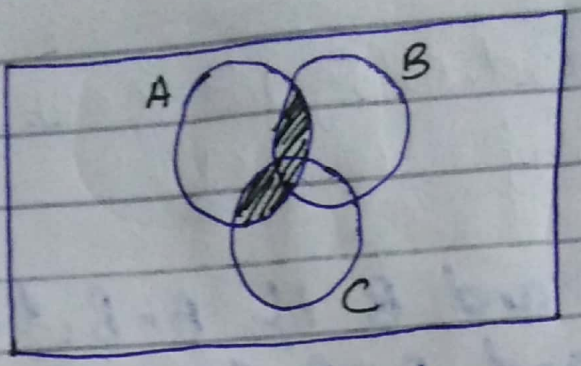
$= \{0, 2, 4, 6\} \cup \{4, 5, 6, 7, 8, 9, 10\}$

$= \{0, 2, 4, 5, 6, 7, 8, 9, 10\}$

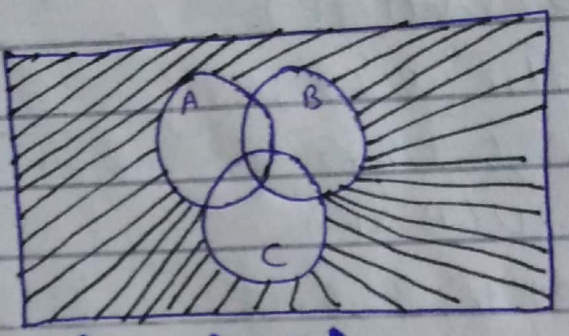
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26. Draw the Venn diagrams for each of these combinations of the sets A, B and C

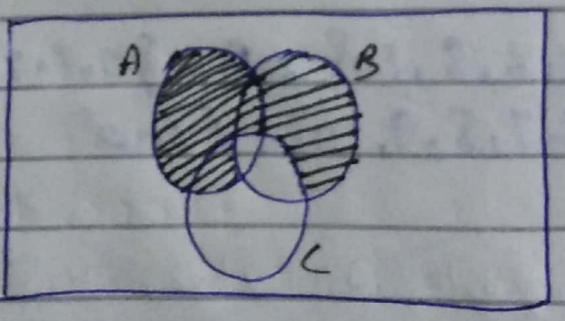
(a) $A \cap (B \cup C)$



(b) $\bar{A} \cap \bar{B} \cap C$

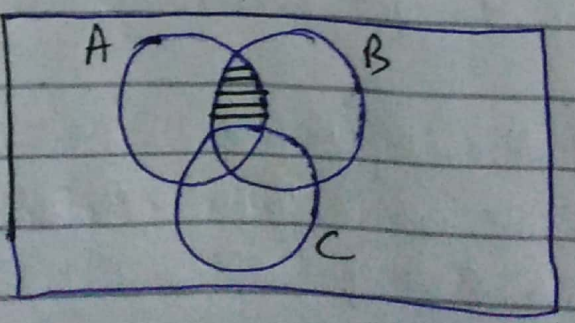


(c) $(A-B) \cup (A-C) \cup (B-C)$



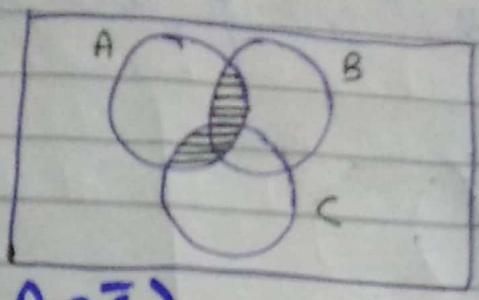
27. Draw the Venn diagrams for each of these combinations of the sets A, B and C.

(a) $A \cap (B-C)$

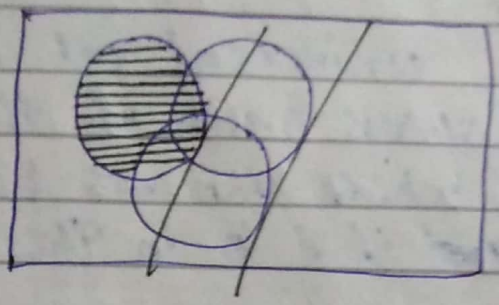


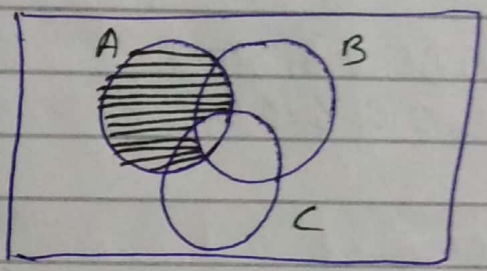
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(b) $(A \cap B) \cup (A \cap C)$



(c) $(A \cap \bar{B}) \cup (A \cap \bar{C})$





32. **Find the symmetric difference of {1,3,5} and {1,2,3}.**

Let $A = \{1,3,5\}$ and $B = \{1,2,3\}$

$A \oplus B = \{1,3,5\} \oplus \{1,2,3\}$

$A \oplus B = \{2,5\}$

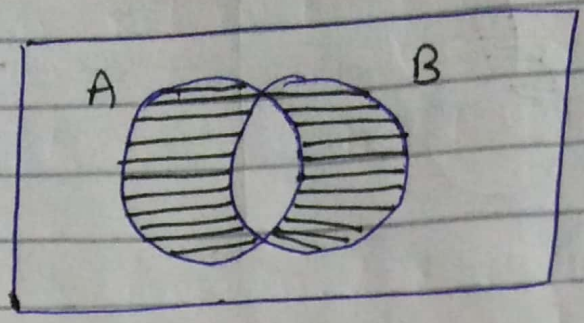
The symmetric difference of A and B, denoted by $A \oplus B$, is the set containing those elements in either A or B, but not in both A and B.

33. **Find the symmetric difference of the set of computer science majors at school and the set of mathematics majors at school**

This will be the set of students majoring in computer science or mathematics but not both.

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34. Draw a Venn diagram for the symmetric difference of the set A and B



52. Suppose that the universal set is $U = \{1,2,3,4,5,6,7,8,9,10\}$. Express each of these sets with bit strings where the ith bit in the bit string is 1 and if i is in the set and 0 otherwise.

$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

$U = 1\ 1\ 1\ 1\ 1\ 1\ 1\ 1\ 1\ 1$

(a) $\{3, 4, 5\}$

0011100000

(b) $\{1, 3, 6, 10\}$

1010010001

(c) $\{2, 3, 4, 7, 8, 9\}$

0111001110

53. Using the same universal set as in the last problem. find the set specified by each of these bit strings.

(a) 11 1100 1111

$\{1, 2, 3, 4, 7, 8, 9, 10\}$

(b) 01 0111 1000

$\{2, 4, 5, 6, 7\}$

(c) 10 0000 0001

$\{1, 10\}$

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59. Find the successor of the following sets.

The successor of the set A is $A \cup \{A\}$

(a) $\{1,2,3\}$

Let $A = \{1,2,3\}$

successor $= A \cup \{A\}$

$= \{1,2,3\} \cup \{\{1,2,3\}\}$

$= \{1,2,3,\{1,2,3\}\}$

(b) ϕ ~~$\{\phi, \{\phi\}\}$~~ $\{\phi\}$

(c) $\{\phi\}$ ~~$\{\{\phi\}, \{\{\phi\}\}\}$~~ $\{\phi, \{\phi\}\}$

(d) $\{\phi, \{\phi\}\}$ $\{\phi, \{\phi\}, \{\phi, \{\phi\}\}\}$

EXERCISE 2.3

1. Why is f not a function from R to R if

(a) $f(x) = 1/x$?

The expression $1/x$ is meaningless for $x=0$, which is one of the elements in domain.

In other words $f(0)$ is not defined.

(b) $f(x) = \sqrt{x}$?

Things like $\sqrt{-3}$ are undefined.

(c) $f(x) = \pm\sqrt{(x^2+1)}$?

We must have $f(x)$ defined uniquely, but there there are 2 values associated with every x, the positive square root and the negative square root of x^2+1.

2. Determine wether f is a function from

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Z to R if

(a) $f(n) = \pm n$?

This is not a function because we do not know that whether $f(3) = 3$ or $f(3) = -3$. For a function, it cannot be both at the same time.

(b) $f(n) = \sqrt{n^2+1}$

This is a function. For all integers n, $\sqrt{n^2+1}$ is a well defined real number.

(c) $f(n) = 1/(n^2-4)$

This is not a function with domain Z because for $n = 2$ and $n = -2$, the function is not defined.

3. Determine whether f is a function from the set of all bit strings to the set of integers if

(a) $f(S)$ is the position of a 0 bit in S.

This is not a function because there may be no 0 bit in S, or there may be more than one 0 bit in S. Thus there may be no value for $f(S)$ or more than one. In either case this violates the definition of function. since function is uniquely defined.

(b) $f(S)$ is the number of 1 bits in S.

This a function since the number of 1 bits is always clearly defined non-negative integer.

(c) $f(S)$ is the smallest integer i such that the ith bit of S is 1 and $f(S) = 0$ when S is the empty string, the string with no bits.

This definition does not tell what to do with a nonempty string consisting of all 0's. Thus, for example, $f(000)$ is undefined.

Therefore, it is not a function

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8. Find these values.

(a) $\lfloor 1.1 \rfloor$

$\lfloor 1.1 \rfloor = 1$

(b) $\lceil 1.1 \rceil$

$\lceil 1.1 \rceil = 2$

(c) $\lfloor -0.1 \rfloor$

$\lfloor -0.1 \rfloor = -1$

(d) $\lceil -0.1 \rceil$

$\lceil -0.1 \rceil = 0$

(e) $\lceil 2.99 \rceil$

$\lceil 2.99 \rceil = 3$

(f) $\lceil -2.99 \rceil$

$\lceil -2.99 \rceil = -2$

(g) $\left\lfloor \frac{1}{2} + \left\lceil \frac{1}{2} \right\rceil \right\rfloor$

$\lfloor 0.5 + \lceil 0.5 \rceil \rfloor = \lfloor 0.5 + 1 \rfloor = \lfloor 1.5 \rfloor = 1$

(h) $\left\lceil \left\lfloor \frac{1}{2} \right\rfloor + \left\lceil \frac{1}{2} \right\rceil + \frac{1}{2} \right\rceil$

$\lceil \lfloor 0.5 \rfloor + \lceil 0.5 \rceil + 0.5 \rceil = \lceil 0 + 1 + 0.5 \rceil = \lceil 1.5 \rceil = 2$

9. Find these values.

(a) $\left\lceil \frac{3}{4} \right\rceil = \lceil 0.75 \rceil = 1$

(b) $\lfloor 7/8 \rfloor = \lfloor 0.875 \rfloor = 0$

(c) $\left\lceil -\frac{3}{4} \right\rceil = \lceil -0.75 \rceil = 0$

(d) $\left\lfloor -\frac{7}{8} \right\rfloor = \lfloor -0.875 \rfloor = -1$

(e) $\lceil 3 \rceil = 3$

(f) $\lfloor -1 \rfloor = -1$

(g) $\left\lfloor \frac{1}{2} + \left\lceil \frac{3}{2} \right\rceil \right\rfloor = \lfloor 0.5 + \lceil 1.5 \rceil \rfloor = \lfloor 0.5 + 2 \rfloor = \lfloor 2.5 \rfloor = 2$

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(h) $\left\lfloor \frac{1}{2} \cdot \left\lfloor \frac{5}{2} \right\rfloor \right\rfloor = \lfloor 0.5 \cdot \lfloor 2.5 \rfloor \rfloor = \lfloor 0.5 \cdot 2 \rfloor = \lfloor 1 \rfloor = 1$

10. Determine whether each of these functions from {a,b,c,d} to itself is one-to-one.

(a) $f(a)=b$, $f(b)=a$, $f(c)=c$, $f(d)=d$

This is one-to-one function.

(b) $f(a)=b$, $f(b)=b$, $f(c)=d$, $f(d)=c$

This not one-to-one function, since b is the image of both a and b.

(c) $f(a)=d$, $f(b)=b$, $f(c)=c$, $f(d)=d$

This not one-to-one function, since d is the image of bothe a and d.

12. Determine whether each of these functions from Z to Z is one-to-one.

(a) $f(n)=n-1$

This is one-to-one function, since if

$$n_1 - 1 = n_2 - 1$$

then $n_1 = n_2$

(b) $f(n)=n^2+1$

This is not one-to-one, since, for example

$f(3)=f(-3)=10$

(c) $f(n)=n^3$

This is one-to-one, since if

$$n_1^3 = n_2^3 \Rightarrow (n_1^3)^{1/3} = (n_2^3)^{1/3}$$

then $n_1 = n_2$

(d) $f(n)=\lceil n/2 \rceil$

This is not one-to-one, since, for example

$f(3)=f(4)=2$

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14. Determine whether $f: \mathbb{Z}\times\mathbb{Z} \to \mathbb{Z}$ is onto if

(a) $f(m,n) = 2m-n$

This is clearly onto, since

$$f(0,-n) = 2(0)-(-n) = 0+n = n$$

for every integer n.

(b) $f(m,n) = m^2-n^2$

This is not onto, since, for example, 2 is not in the range. To see this, if

$$m^2-n^2 = (m-n)(m+n) = 2$$

then m and n must be both even or both odd. In either case, both $m-n$ and $m+n$ are then even, so this expression is divisible by 4 and hence cannot equal 2.

(c) $f(m,n) = m+n+1$

This is onto, since

$$f(0,n-1) = 0+n-1+1 = n$$

for every integer n.

(d) $f(m,n) = |m|-|n|$

This is onto. To achieve negative values we set $m=0$ and to achieve non-negative we set $n=0$

(e) $f(m,n) = m^2-4$

This is not onto because 2 is not in the range.

15. Determine whether the function $f: \mathbb{Z}\times\mathbb{Z} \to \mathbb{Z}$ is onto if

(a) $f(m,n) = m+n$

Given any integer n, we have

$$f(0,n) = 0+n = n$$

So function is onto.

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(b) $f(m,n) = m^2+n^2$

The range contains no negative integers, so the function is not onto.

(c) $f(m,n) = m$

Given any integer m, we have

$$f(m,25) = m$$

so the function is onto.

(We can use any constant in place of n)

(d) $f(m,n) = |n|$

The range contains no negative integers, so the function is not onto.

(e) $f(m,n) = m-n$

Given any integer m, we have

$$f(m,0) = m-0 = m$$

or for any integer n, we have

$$f(0,-n) = 0-(-n) = n$$

So, the function is onto.

22. Determine whether each of these function is a bijection from R to R.

(a) $f(x) = -3x+4$

This is a bijection since the inverse function is $f^{-1}(x) = \frac{4-x}{3}$

(b) $f(x) = -3x^2+7$

This is not one-to-one since $f(17) = f(-17)$, for instance, It is also not onto, since the range is the interval $(-\infty, 7]$.

(c) $f(x) = (x+1)/(x+2)$

This function is not a bijection because $x=-2$ is not in domain and $x=1$ is not in range

OR

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This function is a bijection, but not from R to R because $x=-2$ is not in domain and $x=+1$ On the otherhand, f is a bijection from $R-\{-2\}$ to $R-\{-1\}$, since its inverse is $f^{-1}(x) = (1-2x)/(x-1)$.

(d) $f(x) = x^5+1$

This is a bijection, since its inverse is $f^{-1}(x) = \sqrt[5]{x-1}$

Determine whether each of these function is a bijection from R to R.

(a) $f(x) = 2x+1$

One-to-one function:-

$$2x_1+1 = 2x_2+1$$
$$2x_1 = 2x_2$$
$$x_1 = x_2$$

Onto function:-

let $f(x) = y$

$$2x+1 = y$$
$$2x = y-1$$
$$x = \frac{y-1}{2}$$

put into the function

$$f(x) = 2\left(\frac{y-1}{2}\right)+1$$
$$f(x) = y-1+1$$
$$f(x) = y$$

This is the bijection function.

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(b) $f(x) = x^2 + 1$

One-to-one function

$$x_1^2 + 1 = x_2^2 + 1$$

$$x_1^2 = x_2^2$$

but $x_1 \neq x_2$

but it is not one-to-one

because $f(2) = f(-2) = 5$ so $x_1 \neq x_2$

Onto function

Let $f(x) = y$

$$x^2 + 1 = y$$

$$x^2 = y - 1$$

$$x = \sqrt{y - 1}$$

It is not onto because in range only 1 or greater numbers are included written as $[1, \infty)$

So, it is not bijection.

(c) $f(x) = x^3$

This function is a bijection, since it has inverse function $f(y) = y^{1/3}$

(d) $f(x) = (x^2+1)/(x^2+2)$

This function is not a bijection. It is easy to see that it is not one-to-one because x and $-x$ have same image and range is $[0.5, 1)$ so it is also not onto.

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Exercise 2.4

1. Find these terms of the sequences $\{a_n\}$, where $a_n = 2.(-3)^n + 5^n$

(a) a_0

$a_0 = 2.(-3)^0 + 5^0 = 2.1 + 1 = 3$ $\boxed{a_0 = 3}$

(b) a_1

$a_1 = 2.(-3)^1 + 5^1 = 2.(-3) + 5 = -1$ $\boxed{a_0 = -1}$

(c) a_4

$a_4 = 2.(-3)^4 + 5^4 = 2.(81) + 625 = 787$ $\boxed{a_4 = 787}$

(d) a_5

$a_5 = 2.(-3)^5 + 5^5 = 2.(-243) + 3125 = 2639$ $\boxed{a_5 = 2639}$

2. What is the term a_8 of the sequence $\{a_n\}$ if a_n equals

(a) 2^{n-1}?

$a_8 = 2^{8-1} = 2^7 = 128$ $\boxed{a_8 = 128}$

(b) 7?

$\boxed{a_8 = 7}$

(c) $1 + (-1)^n$?

$a_8 = 1 + (-1)^8 = 1 + 1 = 2$ $\boxed{a_8 = 2}$

(d) $-(-2)^n$?

$a_8 = -(-2)^8 = -256$ $\boxed{a_8 = -256}$

3. What are the terms a_0, a_1, a_2 and a_3 of the

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sequence $\{a_n\}$, where a_n equals

(a) 2^n+1?

$a_0 = 2^0+1 = 1+1 = 2$

$a_1 = 2^1+1 = 2+1 = 3$

$a_2 = 2^2+1 = 4+1 = 5$

$a_3 = 2^3+1 = 8+1 = 9$

$a_0=2 ; a_1=3 ; a_2=5 ; a_3=9$

(b) $(n+1)^{n+1}$?

$a_0 = (0+1)^{0+1} = 1^1 = 1$

$a_1 = (1+1)^{1+1} = 2^2 = 4$

$a_2 = (2+1)^{2+1} = 3^3 = 27$

$a_3 = (3+1)^{3+1} = 4^4 = 256$

$a_0=1 ; a_1=4 ; a_2=27 ; a_3=256$

(c) $\lfloor n/2 \rfloor$?

$a_0 = \lfloor 0/2 \rfloor = \lfloor 0 \rfloor = 0$

$a_1 = \lfloor 1/2 \rfloor = \lfloor 0.5 \rfloor = 0$

$a_2 = \lfloor 2/2 \rfloor = \lfloor 1 \rfloor = 1$

$a_3 = \lfloor 3/2 \rfloor = \lfloor 1.5 \rfloor = 1$

$a_0=0 ; a_1=0 ; a_2=1 ; a_3=1$

(d) $\lfloor n/2 \rfloor + \lceil n/2 \rceil$?

$a_0 = \lfloor 0/2 \rfloor + \lceil 0/2 \rceil = \lfloor 0 \rfloor + \lceil 0 \rceil = 0+0 = 0$

$a_1 = \lfloor 1/2 \rfloor + \lceil 1/2 \rceil = \lfloor 0.5 \rfloor + \lceil 0.5 \rceil = 0+1 = 1$

$a_2 = \lfloor 2/2 \rfloor + \lceil 2/2 \rceil = \lfloor 1 \rfloor + \lceil 1 \rceil = 1+1 = 2$

$a_3 = \lfloor 3/2 \rfloor + \lceil 3/2 \rceil = \lfloor 1.5 \rfloor + \lceil 1.5 \rceil = 1+2 = 3$

$a_0=0 ; a_1=1 ; a_2=2 ; a_3=3$

4. What are the terms a_0, a_1, a_2 and a_3 of the sequence $\{a_n\}$, where a_n equals

(a) $(-2)^n$?

$a_0 = (-2)^0 = 1$

$a_1 = (-2)^1 = -2$

$a_2 = (-2)^2 = 4$

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$a_3 = (-2)^3 = -8$

$a_0 = 1; a_1 = -2; a_2 = 4; a_3 = -8$

(b) 3?

$a_0 = 3; a_1 = 3; a_2 = 3; a_3 = 3$

(c) $7 + 4^n$?

$a_0 = 7 + 4^0 = 7 + 1 = 8$

$a_1 = 7 + 4^1 = 7 + 4 = 11$

$a_2 = 7 + 4^2 = 7 + 16 = 23$

$a_3 = 7 + 4^3 = 7 + 64 = 71$

$a_0 = 8; a_1 = 11; a_2 = 23; a_3 = 71$

(d) $2^n + (-2)^n$?

$a_0 = 2^0 + (-2)^0 = 1 + 1 = 2$

$a_1 = 2^1 + (-2)^1 = 2 - 2 = 0$

$a_2 = 2^2 + (-2)^2 = 4 + 4 = 8$

$a_3 = 2^3 + (-2)^3 = 8 - 8 = 0$

$a_0 = 2; a_1 = 0; a_2 = 8; a_3 = 0$

5. List the first 10 terms of each of these sequences

(a) the sequence that begins with 2 and in which each successive term is 3 more than the preceding term

2, 5, 8, 11, 14, 17, 20, 23, 26, 29

(b) the sequence that lists each positive integer three times in increasing order

1, 1, 1, 2, 2, 2, 3, 3, 3, 4

(c) the sequences that lists the odd positive integers in increasing order, listing each odd integer twice

1, 1, 3, 3, 5, 5, 7, 7, 9, 9

(d) the sequence whose nth term is $n! - 2^n$

0, -1, -2, -2, 8, 88, 656, 4912, 40,064, 362,368, 3627,776

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(e) the sequence that begins with 3, where each succeeding term is twice the preceding term.

3, 6, 12, 24, 48, 96, 192, 384, 768, 1536

(f) the sequence whose first time is 2, second term is 4, and each succeeding term is the sum of the two preceeding terms.

2, 4, 6, 10, 16, 26, 42, 68, 110, 178

(g) the sequence whose n^{th} term is the number of bits in the binary expansion of the number n

1, 2, 2, 3, 3, 3, 3, 4, 4, 4

(h) the sequence where the nth term is the number of letters in English word for the index n

3, 3, 5, 4, 4, 3, 5, 5, 4, 3

6. List the first 10 terms of each of these sequences

(a) the sequence obtained by starting with 10 and obtaining each term by subtracting 3 from the previous term

$a_1 = 10$

$a_2 = a_1 - 3 = 10 - 3 = 7$

$a_3 = a_2 - 3 = 7 - 3 = 4$

$a_4 = a_3 - 3 = 4 - 3 = 1$

$a_5 = a_4 - 3 = 1 - 3 = -2$

$a_6 = a_5 - 3 = -2 - 3 = -5$

$a_7 = a_6 - 3 = -5 - 3 = -8$

$a_8 = a_7 - 3 = -8 - 3 = -11$

$a_9 = a_8 - 3 = -11 - 3 = -14$

$a_{10} = a_9 - 3 = -14 - 3 = -17$

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(b) the sequence whose n^{th} term is the sum of the first n positive integers.

$a_1 = 1$

$a_2 = a_1 + 2 = 1 + 2 = 3$

$a_3 = a_2 + 3 = 3 + 3 = 6$

$a_4 = a_3 + 4 = 6 + 4 = 10$

$a_5 = a_4 + 5 = 10 + 5 = 15$

$a_6 = a_5 + 6 = 15 + 6 = 21$

$a_7 = a_6 + 7 = 21 + 7 = 28$

$a_8 = 28 + 8 = 36$

$a_9 = a_8 + 9 = 36 + 9 = 45$

$a_{10} = a_9 + 10 = 45 + 10 = 55$

(c) the sequence whose n^{th} term is $3^n - 2^n$

$a_1 = 3^1 - 2^1 = 3 - 2 = 1$

$a_2 = 3^2 - 2^2 = 9 - 4 = 5$

$a_3 = 3^3 - 2^3 = 27 - 8 = 19$

$a_4 = 3^4 - 2^4 = 81 - 16 = 65$

$a_5 = 3^5 - 2^5 = 243 - 32 = 211$

$a_6 = 3^6 - 2^6 = 729 - 64 = 665$

$a_7 = 3^7 - 2^7 = 2187 - 128 = 2059$

$a_8 = 3^8 - 2^8 = 6561 - 256 = 6305$

$a_9 = 3^9 - 2^9 = 19683 - 512 = 19171$

$a_{10} = 3^{10} - 2^{10} = 59049 - 1024 = 58025$

(d) the sequence whose n^{th} term is $\lfloor \sqrt{n} \rfloor$

$a_1 = \lfloor \sqrt{1} \rfloor = \lfloor 1 \rfloor = 1$

$a_2 = \lfloor \sqrt{2} \rfloor = \lfloor 1.4 \rfloor = 1$

$a_3 = \lfloor \sqrt{3} \rfloor = \lfloor 1.7 \rfloor = 1$

$a_4 = \lfloor \sqrt{4} \rfloor = \lfloor 2 \rfloor = 2$

$a_5 = \lfloor \sqrt{5} \rfloor = \lfloor 2.2 \rfloor = 2$

$a_6 = \lfloor \sqrt{6} \rfloor = \lfloor 2.4 \rfloor = 2$

$a_7 = \lfloor \sqrt{7} \rfloor = \lfloor 2.6 \rfloor = 2$

$a_8 = \lfloor \sqrt{8} \rfloor = \lfloor 2.8 \rfloor = 2$

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(e) the sequence whose first two terms are 1 and 5 and each succeeding term is the sum of the two previous terms.

$a_1 = 1$

$a_2 = 5$

$a_3 = a_1 + a_2 = 1 + 5 = 6$

$a_4 = a_2 + a_3 = 5 + 6 = 11$

$a_5 = a_3 + a_4 = 6 + 11 = 17$

$a_6 = a_4 + a_5 = 11 + 17 = 28$

$a_7 = a_5 + a_6 = 17 + 28 = 45$

$a_8 = a_6 + a_7 = 28 + 45 = 73$

$a_9 = a_7 + a_8 = 45 + 73 = 118$

$a_{10} = a_8 + a_9 = 73 + 118 = 191$

(f) the sequence whose n^{th} term is the largest integer whose binary expansion has n bits $a_n = 2^n - 1$

The largest integer with a binary expansion with 1 bit is 1 (binary), which corresponds with 1 (decimal). or $a_1 = 2^1 - 1 = 2 - 1 = 1$

$a_1 = 1$

" 2 bit is 11 (binary), which corresponds with 3 (decimal)

$a_2 = 3$ or $a_2 = 2^2 - 1 = 4 - 1 = 3$

" 3 bit is 111 (binary), which corresponds with 7 (decimal)

$a_3 = 7$ or $a_3 = 2^3 - 1 = 8 - 1 = 7$

" 4 bit is 1111 (binary), which corresponds with 15 (decimal)

$a_4 = 15$ or $a_4 = 2^4 - 1 = 16 - 1 = 15$

" 5 bits is 11111 (binary), which corresponds with 31 (decimal)

$a_5 = 31$ or $a_5 = 2^5 - 1 = 32 - 1 = 31$

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" 6 bits is 111111 (binary), which corresponds to 63 (decimal)

$a_6 = 63$ or $a_6 = 2^6 - 1 = 64 - 1 = 63$

" 7 bits is 1111111 (binary), which corresponds to 127 (decimal)

$a_7 = 127$ or $a_7 = 2^7 - 1 = 128 - 1 = 127$

" 8 bits is 11111111 (binary), which corresponds to 255 (decimal).

$a_8 = 255$ or $a_8 = 2^8 - 1 = 256 - 1 = 255$

" 9 bits is 111111111 (binary), which corresponds to 511 (decimal)

$a_9 = 511$ or $a_9 = 2^9 - 1 = 512 - 1 = 511$

" 10 bits is 1111111111 (binary), which corresponds to 1023 (decimal)

$a_{10} = 1023$ or $a_{10} = 2^{10} - 1 = 1024 - 1 = 1023$

(g) the sequence whose terms are constructed sequentially as follows: start with 1, then add 1, then multiply by 1, then add 2, then multiply by 2, and so on.

$a_1 = 1$

$a_2 = a_1 + 1 = 1 + 1 = 2$

$a_3 = a_2 \times 1 = 2 \times 1 = 2$

$a_4 = a_3 + 2 = 2 + 2 = 4$

$a_5 = a_4 \times 2 = 4 \times 2 = 8$

$a_6 = a_5 + 3 = 8 + 3 = 11$

$a_7 = a_6 \times 3 = 11 \times 3 = 33$

$a_8 = a_7 + 4 = 33 + 4 = 37$

$a_9 = a_8 \times 4 = 37 \times 4 = 148$

$a_{10} = a_9 + 5 = 148 + 5 = 153$

(h) the sequence whose n^{th} term is the largest integer k such that $k \leq n$

$a_1 = 1! = 1$

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$a_2 = 2! = 2$

a_3 = Since $2! = 2$ and $3! = 6$

$a_3 = 2$

$a_4 = 2$

$a_5 = 2$

Since $3! = 6$ and $4! = 24$

$a_6 = 3$

$a_7 = 3$

$a_8 = 3$

$a_9 = 3$

$a_{10} = 3$

9. Find the first five terms of the sequence defined by each of these recurrence relations and initial conditions.

(a) $a_n = 6a_{n-1}$, $a_0 = 2$

$a_1 = 6a_{1-1} = 6a_0 = 6 \times 2 = 12$

$a_2 = 6a_{2-1} = 6a_1 = 6 \times 12 = 72$

$a_3 = 6a_{3-1} = 6a_2 = 6 \times 72 = 432$

$a_4 = 6a_{4-1} = 6a_3 = 6 \times 432 = 2592$

$a_5 = 6a_{5-1} = 6a_4 = 6 \times 2592 = 15552$

(b) $a_n = a_{n-1}^2$, $a_0 = 2$

$a_0 = 2$

$a_1 = a_{1-1}^2 = a_0^2 = 2^2 = 4$

$a_2 = a_{2-1}^2 = a_1^2 = 4^2 = 16$

$a_3 = a_{3-1}^2 = a_2^2 = 16^2 = 256$

$a_4 = a_{4-1}^2 = a_3^2 = 256^2 = 65536$

(c) $a_n = a_{n-1} + 3a_{n-2}$, $a_0 = 1$, $a_1 = 2$

$a_0 = 1$

$a_1 = 2$

$a_2 = a_{2-1} + 3a_{2-2}$

$\Rightarrow a_2 = a_1 + 3a_0 = 2 + 3(1) = 5$

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$a_3 = a_{3-1} + 3a_{3-2} = a_2 + 3a_1$

$\Rightarrow a_3 = 5 + 3(2) = 11$

$a_4 = a_{4-1} + 3a_{4-2} = a_3 + 3a_2$

$\Rightarrow a_4 = 11 + 3(5) = 26$

(d) $a_n = na_{n-1} + n^2 a_{n-2}$, $a_0 = 1$, $a_1 = 1$

$a_0 = 1$

$a_1 = 1$

$a_2 = 2a_{2-1} + 2^2 a_{2-2} = 2a_1 + 4a_0$

$\Rightarrow a_2 = 2(1) + 4(1) = 6$

$a_3 = 3a_{3-1} + 3^2 a_{3-2} = 3a_2 + 9a_1$

$\Rightarrow a_3 = 3(6) + 9(1) = 27$

$a_4 = 4a_{4-1} + 4^2 a_{4-2} = 4a_3 + 16a_2$

$\Rightarrow a_4 = 4(27) + 16(6) = 108 + 96 = 204$

(e) $a_n = a_{n-1} + a_{n-3}$, $a_0 = 1$, $a_1 = 2$, $a_2 = 0$

$a_0 = 1$

$a_1 = 2$

$a_2 = 0$

$a_3 = a_{3-1} + a_{3-3} = a_2 + a_0$

$\Rightarrow a_3 = 0 + 1 = 1$

$a_4 = a_{4-1} + a_{4-3} = a_3 + a_1$

$\Rightarrow a_4 = 1 + 2 = 3$

10. Find the first six terms of the sequence defined by each of these recurrence relations and initials conditions.

(a) $a_n = -2a_{n-1}$, $a_0 = -1$

$a_0 = -1$

$a_1 = -2a_{1-1} = -2a_0 = -2(-1) = 2$

$a_2 = -2a_{2-1} = -2a_1 = -2(2) = -4$

$a_3 = -2a_{3-1} = -2a_2 = -2(-4) = 8$

$a_4 = -2a_{4-1} = -2a_3 = -2(8) = -16$

$a_5 = -2a_{5-1} = -2a_4 = -2(-16) = 32$

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(b) $a_n = a_{n-1} - a_{n-2}$, $a_0 = 2$, $a_1 = -1$

$a_0 = 2$

$a_1 = -1$

$a_2 = a_{2-1} - a_{2-2} = a_1 - a_0 = -1 - (2) = 3$

$a_3 = a_{3-1} - a_{3-2} = a_2 - a_1 = -3 - (-1) = -2$

$a_4 = a_{4-1} - a_{4-2} = a_3 - a_2 = -2 - (-3) = 1$

$a_5 = a_{5-1} - a_{5-2} = a_4 - a_3 = 1 - (-2) = 3$

(c) $a_n = 3a_{n-1}^2$, $a_0 = 1$

$a_0 = 1$

$a_1 = 3a_{1-1}^2 = 3a_0^2 = 3(1)^2 = 3$

$a_2 = 3a_{2-1}^2 = 3a_1^2 = 3(3)^2 = 27$

$a_3 = 3a_{3-1}^2 = 3a_2^2 = 3(27)^2 = 2187$

$a_4 = 3a_{4-1}^2 = 3a_3^2 = 3(2187)^2 = 14348907$

$a_5 = 3a_{5-1}^2 = 3a_4^2 = 3(14348907)^2 = 6.167 \times 10^{14}$

(d) $a_n = na_{n-1} + a_{n-2}^2$, $a_0 = -1$, $a_1 = 0$

$a_0 = -1$

$a_1 = 0$

$a_2 = 2a_{2-1} + a_{2-2}^2 = 2a_1 + a_0^2 = 2(0) + (-1)^2 = 1$

$a_3 = 3a_{3-1} + a_{3-2}^2 = 3a_2 + a_1^2 = 3(1) + (0)^2 = 3$

$a_4 = 4a_{4-1} + a_{4-2}^2 = 3a_3 + a_2^2 = 4(3) + (1)^2 = 103$

$a_5 = 5a_{5-1} + a_{5-2}^2 = 5a_4 + a_3^2 = 5(13) + (3)^2 = 74$

(e) $a_n = a_{n-1} - a_{n-2} + a_{n-3}$, $a_0 = 1$, $a_1 = 1$, $a_2 = 2$

$a_0 = 1$

$a_1 = 1$

$a_2 = 2$

$a_3 = a_{3-1} - a_{3-2} + a_{3-3} = a_2 - a_1 + a_0 = 2 - 1 + 1 = 2$

$a_4 = a_{4-1} - a_{4-2} + a_{4-3} = a_3 - a_2 + a_1 = 2 - 2 + 1 = 1$

$a_5 = a_{5-1} - a_{5-2} + a_{5-3} = a_4 - a_3 + a_2 = 1 - 2 + 2 = 1$

11. Let $a_n = 2^n + 5 \cdot 3^n$ for $n = 0, 1, 2, \ldots$

(a) Find a_0, a_1, a_2, a_3, and a_4

$a_0 = 2^0 + 5 \cdot 3^0 = 1 + 5 \cdot 1 = 6$

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$a_1 = 2^1 + 5 \cdot 3^1 = 2 + 5 \cdot 3 = 17$

$a_2 = 2^2 + 5 \cdot 3^2 = 4 + 5 \cdot 9 = 49$

$a_3 = 2^3 + 5 \cdot 3^3 = 8 + 5 \cdot 27 = 143$

$a_4 = 2^4 + 5 \cdot 3^4 = 16 + 5 \cdot 81 = 421$

(b) Show that $a_2 = 5a_1 - 6a_0$, $a_3 = 5a_2 - 6a_1$ and $a_4 = 5a_3 - 6a_2$

Now $a_2 = 5a_1 - 6a_0$

$49 = 5(17) - 6(6) = 85 - 36 = 49$

$49 = 49$ ✓

Now $a_3 = 5a_2 - 6a_1$

$143 = 5(49) - 6(17) = 245 - 102 = 143$

$143 = 143$ ✓

Now $a_4 = 5a_3 - 6a_2$

$421 = 5(143) - 6(49) = 715 - 294 = 421$

$421 = 421$ ✓

Hence it is showed that $a_2 = 5a_1 - 6a_0$, $a_3 = 5a_2 - 6a_1$ and $a_4 = 5a_3 - 6a_2$.

(c) Show that $a_n = 5a_{n-1} - 6a_{n-2}$ for all all integers ~~to~~ n with $n \geq 2$

As in part (b), we have taken some integers like 2, 3 and 4 and it is showed that these are true / satisfies the equation $a_n = 5a_{n-1} - 6a_{n-2}$

So it is showed that $a_n = 5a_{n-1} - 6a_{n-2}$ for all integers n with $n \geq 2$

12. Show that the sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = -3a_{n-1} + 4a_{n-2}$ if

(a) $a_n = 0$

$a_n = -3(0) + 4(0) = 0$ ✓ $a_n = 0$

(b) $a_n = 1$

$a_n = -3(1) + 4(1) = -3 + 4 = 1$ ✓ $a_n = 1$

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(c) $a_n = (-4)^n$

$a_n = -3(-4)^{n-1} + 4(-4)^{n-2}$

$a_n = -3(-4)^n \cdot (-4)^{-1} + 4(-4)^n \cdot (-4)^{-2}$

$a_n = \frac{-3}{-4}(-4)^n + \frac{4}{(-4)^2}(-4)^n$

$a_n = \frac{3}{4}(-4)^n + \frac{1}{4}(-4)^n$

$a_n = \frac{3+1}{4}(-4)^n$

$a_n = \frac{4}{4}(-4)^n = (-4)^n$ ✓

$a_n = (-4)^n$ proved.

(d) $a_n = 2(-4)^n + 3$

$a_n = -3[+2(-4)^{n-1} + 3] + 4[2(-4)^{n-2} + 3]$

$a_n = -6(-4)^{n-1} - 9 + 8(-4)^{n-2} + 12$

$a_n = -6(-4)^n \cdot (-4)^{-1} + 8(-4)^n \cdot (-4)^{-2} + 3$

$a_n = \frac{-6}{(-4)}(-4)^n + \frac{8}{(-4)^2}(-4)^n + 3$

$a_n = +\frac{3}{2}(-4)^n + \frac{1}{2}(-4)^n + 3$

$a_n = \frac{3+1}{2}(-4)^n + 3$

$a_n = 2(-4)^n + 3$ proved ✓

13. Is the sequence $\{a_n\}$ a solution of the recurrence relation $a_n = 8a_{n-1} - 16a_{n-2}$ if

(a) $a_n = 0$?

$a_n = 8(0) - 16(0) = 0 - 0 = 0$

Yes

(b) $a_n = 1$?

$a_n = 8(1) - 16(1) = 8 - 16 = -8$

No $\because a_n = 1$

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(c) $a_n = 2^n$?

$a_n = 8(2)^{n-1} - 16(2)^{n-2}$

$a_n = 8(2)^n \cdot (2)^{-1} - 16(2)^n \cdot (2)^{-2}$

$a_n = \frac{8}{2}(2)^n - \frac{16}{2^2}(2)^n$

$a_n = 4(2)^n - 4(2)^n$

$a_n = 0$

No $\because a_n = 2^n$

(d) $a_n = 4^n$?

$a_n = 8(4)^{n-1} - 16(4)^{n-2}$

$a_n = 8(4)^n \cdot (4)^{-1} - 16(4)^n \cdot (4)^{-2}$

$a_n = \frac{8}{4}(4)^n - \frac{16}{4^2}(4)^n$

$a_n = 2(4)^n - 1(4)^n$

$a_n = 4^n$

Yes

(e) $a_n = n4^n$?

$a_n = 8[(n-1)(4)^{n-1}] - 16[(n-2)(4)^{n-2}]$

$a_n = (8n-8)(4)^n \cdot (4)^{-1} - (16n+32)(4)^n \cdot (4)^{-2}$

$a_n = \frac{(8n-8)}{4}(4)^n - \frac{(16n-32)}{4^2}(4)^n$

$a_n = (2n-2)(4)^n - (n-2)(4)^n$

$a_n = 2n(4)^n - 2(4)^n - n(4)^n + 2(4)^n$

$a_n = n4^n$

Yes

(f) $a_n = 2.4^n + 3n4^n$?

$a_n = 8[2(4)^{n-1} + 3(n-1)(4)^{n-1}] - 16[2(4)^{n-2} + 3(n-2)(4)^{n-2}]$

$a_n = 8(4)^{n-1}[2 + 3n - 3] - 16(4)^{n-2}[2 + 3n - 6]$

$a_n = 8(4)^n \cdot (4)^{-1}(3n-1) - 16(4)^n \cdot (4)^{-2}(3n-4)$

$a_n = \frac{8}{4}(4)^n(3n-1) - \frac{16}{4^2}(4)^n(3n-4)$

$a_n = 2(3n-1)(4)^n - (4)^n(3n-4)$

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$a_n = 6n4^n - 2.4^n - 3n4^n + 4.4^n$

$a_n = 2.4^n + 3n4^n$

Yes

(g) $a_n = (-4)^n$?

$a_n = 8(-4)^{n-1} - 16(-4)^{n-2}$

$a_n = 8(-4)^n.(-4)^{-1} - 16(-4)^n.(-4)^{-2}$

$a_n = \frac{8}{-4}(-4)^n - \frac{16}{4^2}(-4)^n$

$a_n = -2(-4)^n - (-4)^n$

$a_n = -3(-4)^n$

No $\because a_n = (-4)^n$

(h) $a_n = n^2 4^n$?

$a_n = 8[(n-1)^2(4)^{n-1}] - 16[(n-2)^2(4)^{n-2}]$

$a_n = 8(n^2-2n+1)(4)^n.(4)^{-1} - 16(n^2-4n+4)(4)^n.(4)^{-2}$

$a_n = \frac{8}{4}(n^2-2n+1)(4)^n - \frac{16}{4^2}(n^2-4n+4)(4)^n$

$a_n = 2(n^2-2n+1)(4)^n - (n^2-4n+4)(4)^n$

$a_n = 2n^2 4^n - 4n4^n + 2.4^n - n^2 4^n + 4n4^n - 4.(4)^n$

$a_n = n^2 4^n - 2.4^n$

No $\because a_n = n^2 4^n$

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15. Show that the sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = a_{n-1} + 2a_{n-2} + 2n - 9$ if

(a) $a_n = -n + 2$

$a_n = -(n-1) + 2 + 2[-(n-2) + 2] + 2n - 9$

$a_n = -n + 1 + 2 + 2(-n + 2 + 2) + 2n - 9$

$a_n = n + 3 + 2(-n + 4) - 9$

$a_n = n - 6 - 2n + 8$

$a_n = -n + 2$ ✓ proved.

(b) $a_n = 5(-1)^n - n + 2$

$a_n = 5(-1)^{n-1} - (n-1) + 2 + 2[(5)(-1)^{n-2} - (n-2) + 2] + 2n - 9$

$a_n = 5(-1)^n \cdot (-1)^{-1} - n + 1 + 2 + 2[5(-1)^n (-1)^{-2} - n + 2 + 2] + 2n - 9$

$a_n = \frac{5}{-1}(-1)^n - n + 3 + 2\left[\frac{5}{(-1)^2}(-1)^n - n + 4\right] + 2n - 9$

$a_n = -5(-1)^n - n + 3 + 2(5(-1)^n - n + 4) + 2n - 9$

$a_n = -5(-1)^n - n + 3 + 10(-1)^n - 2n + 8 + 2n - 9$

$a_n = 5(-1)^n - n + 2$ ✓ proved

(c) $a_n = 3(-1)^n + 2^n - n + 2$

$a_n = (3)(-1)^{n-1} + 2^{n-1} - (n-1) + 2 + 2\left[3(-1)^{n-2} + 2^{n-2} - (n-2) + 2\right] + 2n - 9$

$a_n = 3(-1)^n \cdot (-1)^{-1} + (2)^n (2)^{-1} - n + 1 + 2 + 2[3(-1)^n (-1)^{-2} + 2^n (2)^{-2} - n + 2 + 2] + 2n - 9$

$a_n = \frac{3}{-1}(-1)^n + \frac{1}{2}(2)^n - n + 3 + 2\left[\frac{3}{(-1)^2}(-1)^n + \frac{1}{(2)^2}2^n - n + 4\right] + 2n - 9$

$a_n = -3(-1)^n + \frac{1}{2}(2)^n - n + 3 + 2\left(3(-1)^n + \frac{1}{4}(2)^n - n + 4\right) + 2n - 9$

$a_n = -3(-1)^n + \frac{1}{2}(2)^n - n + 3 + 6(-1)^n + \frac{1}{2}(2)^n - 2n + 8 + 2n - 9$

$a_n = 3(-1)^n + 2^n - n + 2$ ✓ proved.

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(d) $a_n = 7.2^n - n + 2$

$a_n = 7.2^{n-1} - (n-1) + 2 + 2[7.2^{n-2} - (n-2) + 2] + 2n - 9$

$a_n = 7.2^n.2^{-1} - n + 1 + 2 + 2[7.2^n.2^{-2} - n + 2 + 2] + 2n - 9$

$a_n = \frac{7}{2}.2^n - n + 3 + 2\left[\frac{7}{4}2^n - n + 4\right] + 2n - 9$

$a_n = \frac{7}{2}.2^n - n + 3 + \frac{7}{2}2^n - 2n + 8 + 2n - 9$

$a_n = 7.2^n - n + 2$ ✓ proved

29. What are the values of these sums?

(a) $\sum_{k=1}^{5} (k+1)$

$\sum_{k=1}^{5} (k+1) = (1+1) + (2+1) + (3+1) + (4+1) + (5+1)$

$= 2 + 3 + 4 + 5 + 6$

$= 20$

(b) $\sum_{j=0}^{4} (-2)^j$

$\sum_{j=0}^{4} (-2)^j = (-2)^0 + (-2)^1 + (-2)^2 + (-2)^3 + (-2)^4$

$= 1 - 2 + 4 - 8 + 16$

$= 11$

(c) $\sum_{i=1}^{10} 3$

$\sum_{i=1}^{10} 3 = 3 \times 10 = 30$

(d) $\sum_{j=0}^{8} (2^{j+1} - 2^j)$

$\sum_{j=0}^{8} (2^{j+1} - 2^j)$

$= (2^{0+1} - 2^0) + (2^{1+1} - 2^1) + (2^{2+1} - 2^2) + (2^{3+1} - 2^3) + (2^{4+1} - 2^4)$
$+ (2^{5+1} - 2^5) + (2^{6+1} - 2^6) + (2^{7+1} - 2^7) + (2^{8+1} - 2^8)$

$= (2^1 - 2^0) + (2^2 - 2^1) + (2^3 - 2^2) + (2^4 - 2^3) + (2^5 - 2^4)$
$+ (2^6 - 2^5) + (2^7 - 2^6) + (2^8 - 2^7) + (2^9 - 2^8)$

$= (2 - 1) + (4 - 2) + (8 - 4) + (16 - 8) + (32 - 16)$
$+ (64 - 32) + (128 - 64) + (256 - 128) + (512 - 256)$

$= 1 + 2 + 4 + 8 + 16 + 32 + 64 + 128 + 256$

$= 511$

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30. What are the values of these sums, where $S = \{1, 3, 5, 7\}$?

(a) $\sum_{j \in S} j$

$$\sum_{j \in S} j = 1 + 3 + 5 + 7 = 16$$

(b) $\sum_{j \in S} j^2$

$$\sum_{j \in S} j^2 = (1)^2 + (3)^2 + (5)^2 + (7)^2$$
$$= 1 + 9 + 25 + 49$$
$$= 84$$

(c) $\sum_{j \in S} (1/j)$

$$\sum_{j \in S} (1/j) = (1/1) + (1/3) + (1/5) + (1/7)$$
$$= 1 + \frac{1}{3} + \frac{1}{5} + \frac{1}{7}$$
$$= \frac{105 + 35 + 21 + 15}{105}$$
$$\sum_{j \in S} (1/j) = \frac{176}{105}$$

(d) $\sum_{j \in S} 1$

$$\sum_{j \in S} 1 = 1 \times 4 = 4 \quad \because \text{no. of elements} = 4$$

31. What is the value of each of these sums of terms of a geometric progression?

(a) $\sum_{j=0}^{8} 3 \cdot 2^j$

$$\sum_{j=0}^{8} 3 \cdot 2^j = 3 \cdot 2^0 + 3 \cdot 2^1 + 3 \cdot 2^2 + 3 \cdot 2^3 + 3 \cdot 2^4 + 3 \cdot 2^5 + 3 \cdot 2^6 + 3 \cdot 2^7 + 3 \cdot 2^8$$
$$= 3 \cdot 1 + 3 \cdot 2 + 3 \cdot 4 + 3 \cdot 8 + 3 \cdot 16 + 3 \cdot 32 + 3 \cdot 64 + 3 \cdot 128 + 3 \cdot 256$$
$$= 3 + 6 + 12 + 24 + 48 + 96 + 192 + 384 + 768$$
$$= 1533$$

(b) $\sum_{j=1}^{8} 2^j$

$$\sum_{j=1}^{8} 2^j = 2^1 + 2^2 + 2^3 + 2^4 + 2^5 + 2^6 + 2^7 + 2^8$$
$$= 2 + 4 + 8 + 16 + 32 + 64 + 128 + 256$$
$$= 510$$

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(c) $\sum_{j=2}^{8} (-3)^j$

$\sum_{j=2}^{8} (-3)^j = (-3)^2 + (-3)^3 + (-3)^4 + (-3)^5 + (-3)^6 + (-3)^7 + (-3)^8$

$= 9 - 27 + 81 - 243 + 729 - 2187 + 6561$

$= 4923$

(d) $\sum_{j=0}^{8} 2 \cdot (-3)^j$

$\sum_{j=0}^{8} 2 \cdot (-3)^j = 2 \cdot (-3)^0 + 2 \cdot (-3)^1 + 2 \cdot (-3)^2 + 2 \cdot (-3)^3 + 2 \cdot (-3)^4 + 2 \cdot (-3)^5 + 2 \cdot (-3)^6 + 2 \cdot (-3)^7 + 2 \cdot (-3)^8$

$= 2 \cdot 1 + 2 \cdot -3 + 2 \cdot 9 + 2 \cdot -27 + 2 \cdot 81 + 2 \cdot -243 + 2 \cdot 729 + 2 \cdot -2187 + 2 \cdot 6561$

$= 2 - 6 + 18 - 54 + 162 - 486 + 1458 - 4374 + 13122$

$= 9842$

32. Find the value of each of these sums.

(a) $\sum_{j=0}^{8} (1 + (-1)^j)$

$= (1 + (-1)^0) + (1 + (-1)^1) + (1 + (-1)^2) + (1 + (-1)^3) + (1 + (-1)^4) + (1 + (-1)^5) + (1 + (-1)^6) + (1 + (-1)^7) + (1 + (-1)^8)$

$= (1+1) + (1-1) + (1+1) + (1-1) + (1+1) + (1-1) + (1+1) + (1-1) + (1+1)$

$= 2 + 0 + 2 + 0 + 2 + 0 + 2 + 0 + 2 = 10$

(b) $\sum_{j=0}^{8} (3^j - 2^j)$

$= (3^0 - 2^0) + (3^1 - 2^1) + (3^2 - 2^2) + (3^3 - 2^3) + (3^4 - 2^4) + (3^5 - 2^5) + (3^6 - 2^6) + (3^7 - 2^7) + (3^8 - 2^8)$

$= (1-1) + (3-2) + (9-4) + (27-8) + (81-16) + (243-32) + (729-64) + (2187-128) + (6561-256)$

$= 0 + 1 + 5 + 19 + 65 + 211 + 665 + 2059 + 6305$

$= 9330$

(c) $\sum_{j=0}^{8} (2 \cdot 3^j + 3 \cdot 2^j)$

$= (2 \cdot 3^0 + 3 \cdot 2^0) + (2 \cdot 3^1 + 3 \cdot 2^1) + (2 \cdot 3^2 + 3 \cdot 2^2) + (2 \cdot 3^3 + 3 \cdot 2^3) + (2 \cdot 3^4 + 3 \cdot 2^4) + (2 \cdot 3^5 + 3 \cdot 2^5) + (2 \cdot 3^6 + 3 \cdot 2^6) + (2 \cdot 3^7 + 3 \cdot 2^7) + (2 \cdot 3^8 + 3 \cdot 2^8)$

$= (2 \cdot 1 + 3 \cdot 1) + (2 \cdot 3) + (3 \cdot 2) + (2 \cdot 9 + 3 \cdot 4) + (2 \cdot 27 + 3 \cdot 8) + (2 \cdot 81 + 3 \cdot 16) + (2 \cdot 243 + 3 \cdot 32) + (2 \cdot 729 + 3 \cdot 64) + (2 \cdot 2187 + 3 \cdot 128) + (2 \cdot 6561 - 3 \cdot 256)$

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$= 2+3+6+6+18+12+54+24+162+48+486+96$
$+1458+192+4374+384+13122+768$
$= 21215$

(d) $\sum_{j=0}^{8}(2^{j+1}-2^j)$
$= (2^{0+1}-2^0)+(2^{1+1}-2^1)+(2^{2+1}-2^2)+(2^{3+1}-2^3)+(2^{4+1}-2^4)$
$+(2^{5+1}-2^5)+(2^{6+1}-2^6)+(2^{7+1}-2^7)+(2^{8+1}-2^8)$
$= (2^1-2^0)+(2^2-2^1)+(2^3-2^2)+(2^4-2^3)+(2^5-2^4)$
$+(2^6-2^5)+(2^7-2^6)+(2^8-2^7)+(2^9-2^8)$
$= (2-1)+(4-2)+(8-4)+(16-8)+(32-16)+(64-32)+(128-64)$
$+(256-128)+(512-256)$
$= 1+2+4+8+16+32+64+128+256 = 511$

33. Compute each of these double sums.

(a) $\sum_{i=1}^{2}\sum_{j=1}^{3}(i+j)$
$= \sum_{i=1}^{2}\left((i+1)+(i+2)+(i+3)\right)$ or $\sum_{i=1}^{2}(3i+6)$
or $(3(0)+6)+(3(1)+6)+(3(2)+6) = 21$
$= \left((1+1)+(2+1)\right)+\left((1+2)+(2+2)\right)+\left((1+3)+(2+3)\right)$
$= 2+3+3+4+4+5$
$= 21$

(b) $\sum_{i=0}^{2}\sum_{j=0}^{3}(2i+3j)$
$= \sum_{i=0}^{2}\left((2i+3(0))+(2i+3(1))+(2i+3(2))+(2i+3(3))\right)$
$= \sum_{i=0}^{2}(2i+0+2i+3+2i+6+2i+9)$
$= \sum_{i=0}^{2}(8i+18)$
$= 8(0)+18+8(1)+18+8(2)+18$
$= 18+8+18+16+18 = 78$

(c) $\sum_{i=0}^{3}\sum_{j=0}^{2}i$

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(c) $\sum_{i=1}^{3}\sum_{j=0}^{2} i$

$= \sum_{i=1}^{3}(i+i+i) = \sum_{i=1}^{3} 3i$

$= 3\sum_{i=1}^{3} i$

$= 3(1+2+3) = 3(6)$

$= 18$

(d) $\sum_{i=0}^{2}\sum_{j=1}^{3} ij$

$= \sum_{i=0}^{2}(i+2i+3i)$

$= \sum_{i=0}^{2}(6i)$

$= 6(0)+6(1)+6(2) = 6+12$

$= 18$

34. Compute each of these double sums

(a) $\sum_{i=1}^{3}\sum_{j=1}^{2}(i-j)$

$= \sum_{i=1}^{3}((i-1)+(i-2))$

$= \sum_{i=1}^{3}(2i-3)$

$= 2(1)-3+2(2)-3+2(3)-3$

$= 2-3+4-3+6-3$

$= 3$

(b) $\sum_{i=0}^{3}\sum_{j=0}^{2}(3i+2j)$

$= \sum_{i=0}^{3}[(3i+2(0))+(3i+2(1))+(3i+2(2))]$

$= \sum_{i=0}^{3}(3i+0+3i+2+3i+4)$

$= \sum_{i=0}^{3}(9i+6)$

$= 9(0)+6+9(1)+6+9(2)+6+9(3)+6$

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$= 6+9+6+18+6+27+6$

$= 78$

(b) $\sum_{i=0}^{3}\sum_{j=0}^{2} j$

$= \sum_{i=0}^{3} (j+j+j) = \sum_{i=0}^{3} 4j$

$= 3\sum_{i=0}^{3} j$

$= 3(0+1+2+3)$

$= 3(6) = 18$ X

(c) $\sum_{i=0}^{3}\sum_{j=0}^{2} j$

$= \sum_{i=0}^{3} (0+1+2)$

$= \sum_{i=0}^{3} 3$

$= 3 \times 3 = 9$

or $= 3+3+3 = 9$

(d) $\sum_{i=0}^{2}\sum_{j=0}^{3} i^2 j^3$

$= \sum_{i=0}^{2} (i^2(0)^3 + i^2(1)^3 + i^2(2)^3 + i^2(3)^3)$

$= \sum_{i=0}^{2} (0 + i^2 + 8i^2 + 27i^2)$

$= \sum_{i=0}^{2} 36i^2 = 36\sum_{i=0}^{2} i^2$

$= 36(0^2+1^2+2^2) = 36(1+4)$

$= 36(5) = 180$

Examples From Notes

1. Find the 20th term of A.S

3, 9, 15, 21, ...

$a_1 = 3$, $d = 9-3 = 6$, $n = 20$

$a_n = a_1 + (n-1)d$

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$a_{20} = 3 + (20-1)6$

$a_{20} = 3 + (19)(6)$

$a_{20} = 3 + 114$

$\boxed{a_{20} = 117}$

2. Which term of A.S $4, 1, -2, \ldots\ldots, -77$

$a_1 = 4$, $a_n = -77$, $d = 1 - 4 = -3$

$a_n = a_1 + (n-1)d$

$-77 = 4 + (n-1)(-3)$

$-77 = 4 - 3n + 3$

$-77 = 7 - 3n$

$-3n = -77 - 7$

$-3n = -84$

$n = \frac{-84}{3}$

$\boxed{n = 28}$

3. Find the 36th term of A.S when 3rd term is 7 and 8th term is 17.

$a_3 = a_1 + (3-1)d$

$a_3 = a_1 + 2d$ — (1)

$a_8 = a_1 + (8-1)d$

$a_8 = a_1 + 7d$ — (2)

& Subtracting (1) from (2)

$a_1 + 7d = 17$

$\pm a_1 \pm 2d = \pm 7$

$5d = 10$

$d = 10/5$

$\boxed{d = 2}$

putting in (1)

$a_1 + 2(2) = 7$

$a_1 + 4 = 7$

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$a_1 = 7 - 4 = 3$

New $a_{36} = a_1 + (36-1)d$

$a_{36} = 3 + (35)(2)$

$a_{36} = 3 + 70$

$\boxed{a_{36} = 73}$

Note:-

$a_n = a_1 r^{n-1}$

where $r = \frac{a_2}{a_1}$

4. Find the 8th term of G.S

4, 12, 36, ----

$a_1 = 4$

$r = \frac{12}{4} = 3$, $n = 8$

$a_n = a_1 r^{n-1}$

$a_8 = 4(3)^{8-1}$

$a_8 = 4(3)^7$

$a_8 = 4(2187)$

$\boxed{a_8 = 8748}$

5. Which term of G.S is 1/8 if first term is 4 and $r = 1/2$

$a_n = 1/8$

$a_1 = 4$

$r = 1/2$

$a_n = a_1 r^{n-1}$

$\left(\frac{1}{8}\right) = 4\left(\frac{1}{2}\right)^{n-1}$

$\left(\frac{1}{2}\right)^{n-1} = \frac{1}{32}$

$\left(\frac{1}{2}\right)^{n-1} = \left(\frac{1}{2}\right)^5$

$n - 1 = 5$

$n = 5 + 1$

$\boxed{n = 6}$

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6. Write the G.S with posite term whose 2nd term is 9 and 4th term is 1.

$$a_2 = a_1 r^{2-1}$$

$$9 = a_1 r \quad \text{—} (1)$$

$$a_4 = a_1 r^{4-1}$$

$$1 = a_1 r^3 \quad \text{—} (2)$$

Dividing equation (2) by (1)

$$\frac{1}{9} = \frac{a_1 r^3}{a_1 r}$$

$$\frac{1}{9} = r^2$$

$$r = \pm \frac{1}{3}$$

negative negative

$$r = \frac{1}{3}$$

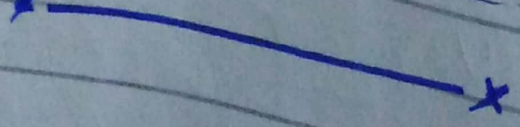
putting in (1) $r = 1/3$

$$9 = a_1 (1/3)$$

$$a_1 = 27$$

G.S is

$$27, 9, 3, 1, \ldots$$



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EXERCISE 2.6

1. Let $A = \begin{bmatrix} 1 & 1 & 1 & 3 \\ 2 & 0 & 4 & 6 \\ 1 & 1 & 3 & 7 \end{bmatrix}$

(a) What size is A?

No. of rows = 3

No. of columns = 4

Size of A = 3 × 4

(b) What is the third column of A?

The third column of A is 3×1 matrix

$$\begin{bmatrix} 1 \\ 4 \\ 3 \end{bmatrix}$$

(c) What is the second row of A?

The second row of A is 1×4 matrix

$$\begin{bmatrix} 2 & 0 & 4 & 6 \end{bmatrix}$$

(d) What is the element of A in the (3,2)th position?

This is the element in 3rd row and 2nd column of A that is 1.

(e) What is A^t?

$$A^t = \begin{bmatrix} 1 & 1 & 1 & 3 \\ 2 & 0 & 4 & 6 \\ 1 & 1 & 3 & 7 \end{bmatrix}$$

$$A^t = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 0 & 1 \\ 1 & 4 & 3 \\ 3 & 6 & 7 \end{bmatrix}$$

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2. **Find A+B, where**

(a) $A = \begin{bmatrix} 1 & 0 & 4 \\ -1 & 2 & 2 \\ 0 & -2 & -3 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 3 & 5 \\ 2 & 2 & -3 \\ 2 & -3 & 0 \end{bmatrix}$

$$A+B = \begin{bmatrix} 1-1 & 0+3 & 4+5 \\ -1+2 & 2+2 & 2-3 \\ 0+2 & -2-3 & -3+0 \end{bmatrix}$$

$$A+B = \begin{bmatrix} 0 & 3 & 9 \\ 1 & 4 & -1 \\ 2 & -5 & -3 \end{bmatrix}$$

(b) $A = \begin{bmatrix} -1 & 0 & 5 & 6 \\ -4 & -3 & 5 & -2 \end{bmatrix}$, $B = \begin{bmatrix} -3 & 9 & -3 & 4 \\ 0 & -2 & -1 & 2 \end{bmatrix}$

$$A+B = \begin{bmatrix} -1-3 & 0+9 & 5-3 & 6+4 \\ -4+0 & -3-2 & 5-1 & -2+2 \end{bmatrix}$$

$$A+B = \begin{bmatrix} -4 & 9 & 2 & 10 \\ -4 & -5 & 4 & 0 \end{bmatrix}$$

3. **Find AB, if**

(a) $A = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 4 \\ 1 & 3 \end{bmatrix}$

$$AB = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 0 & 4 \\ 1 & 3 \end{bmatrix}$$

$$AB = \begin{bmatrix} (2\times0)+(1\times1) & (2\times4)+(1\times3) \\ (3\times0)+(2\times1) & (3\times4)+(2\times3) \end{bmatrix}$$

$$AB = \begin{bmatrix} 0+1 & 8+3 \\ 0+2 & 12+6 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 & 11 \\ 2 & 18 \end{bmatrix}$$

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(b) $A_2 \begin{bmatrix} 1 & -1 \\ 0 & 1 \\ 2 & 3 \end{bmatrix}$, $B_2 \begin{bmatrix} 3 & -2 & -1 \\ 1 & 0 & 2 \end{bmatrix}$

$$AB_2 \begin{bmatrix} (1\times3)+(-1\times1) & (1\times-2)+(-1\times0) & (1\times-1)+(-1\times2) \\ (0\times3)+(1\times1) & (0\times-2)+(1\times0) & (0\times-1)+(1\times2) \\ (2\times3)+(3\times1) & (2\times-2)+(3\times0) & (2\times-1)+(3\times2) \end{bmatrix}$$

$$AB_2 \begin{bmatrix} 3+1 & -2+0 & -1-2 \\ 0+1 & 0+0 & 0+2 \\ 6+3 & -4+0 & -2+6 \end{bmatrix}$$

$$AB_2 \begin{bmatrix} 2 & -2 & -3 \\ 1 & 0 & 2 \\ 9 & -4 & 4 \end{bmatrix}$$

(c) $A_2 \begin{bmatrix} 4 & -3 \\ 3 & -1 \\ 0 & -2 \\ -1 & 5 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 3 & 2 & -2 \\ 0 & -1 & 4 & -3 \end{bmatrix}$

$$AB_2 \begin{bmatrix} (4\times(-1))+(-3\times0) & (4\times3)+(-3\times-1) & (4\times2)+(-3\times4) & (4\times-2)+(-3\times-3) \\ (3\times-1)+(-1\times0) & (3\times3)+(-1\times-1) & (3\times2)+(-1\times4) & (3\times-2)+(-1\times-3) \\ (0\times-1)+(-2\times0) & (0\times3)+(-2\times-1) & (0\times2)+(-2\times4) & (0\times-2)+(-2\times-3) \\ (-1\times-1)+(5\times0) & (-1\times3)+(5\times-1) & (-1\times2)+(5\times4) & (-1\times-2)+(5\times-3) \end{bmatrix}$$

$$AB_2 \begin{bmatrix} -4+0 & 12+3 & 8-12 & -8+9 \\ -3+0 & 9+1 & 6-4 & -6+3 \\ 0+0 & 0+2 & 0-8 & 0+(-6) \\ 1+0 & -3-5 & -2+20 & 2-15 \end{bmatrix}$$

$$AB_2 \begin{bmatrix} -4 & 15 & -4 & 1 \\ -3 & 10 & 2 & -3 \\ 0 & 2 & -8 & -6 \\ 1 & -8 & +18 & -13 \end{bmatrix}$$

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4. Find the product AB, where

(a) $A_2 \begin{bmatrix} 1 & 0 & 1 \\ 0 & -1 & -1 \\ -1 & 1 & 0 \end{bmatrix}$, $B_2 \begin{bmatrix} 0 & 1 & -1 \\ 1 & -1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$

$$AB_2 \begin{bmatrix} (1\times0)+(0\times1)+(1\times-1) & (1\times1)+(0\times-1)+(1\times0) & (1\times-1)+(0\times0)+(1\times1) \\ (0\times0)+(-1\times1)+(-1\times-1) & (0\times1)+(-1\times-1)+(-1\times0) & (0\times-1)+(-1\times0)+(-1\times1) \\ (-1\times0)+(1\times1)+(0\times-1) & (-1\times1)+(1\times-1)+(0\times0) & (-1\times-1)+(1\times0)+(0\times1) \end{bmatrix}$$

$$AB_2 \begin{bmatrix} 0+0-1 & 1+0+0 & -1+0+1 \\ 0-1+1 & 0+1+0 & 0+0-1 \\ 0+1+0 & -1-1+0 & 1+0+0 \end{bmatrix}$$

$$AB_2 \begin{bmatrix} -1 & 1 & 0 \\ 0 & 1 & -1 \\ 1 & -2 & 1 \end{bmatrix}$$

(b) $A_2 \begin{bmatrix} 1 & -3 & 0 \\ 1 & 2 & 2 \\ 2 & 1 & -1 \end{bmatrix}$, $B_2 \begin{bmatrix} 1 & -1 & 2 & 3 \\ -1 & 0 & 3 & -1 \\ -3 & -2 & 0 & 2 \end{bmatrix}$

5.

$$AB_2 \begin{bmatrix} (1\times1)+(-3\times-1)+(0\times-3) & (1\times-1)+(-3\times0)+(0\times-2) & (1\times2)+(-3\times3)+(0\times0) & (1\times3)+(-3\times-1)+(0\times2) \\ (1\times1)+(2\times-1)+(2\times-3) & (1\times-1)+(2\times0)+(2\times-2) & (1\times2)+(2\times3)+(2\times0) & (1\times3)+(2\times-1)+(2\times-2) \\ (2\times1)+(1\times-1)+(-1\times-3) & (2\times-1)+(1\times0)+(-1\times-2) & (2\times2)+(1\times3)+(-1\times0) & (2\times3)+(1\times-1)+(-1\times-2) \end{bmatrix}$$

$$AB_2 \begin{bmatrix} 1+3+0 & -1+0+0 & 2-9 & 3+3 \\ 1-2-6 & -1+0-4 & 2+6 & 3-2+4 \\ 2-1+3 & -2+0+2 & 4+3 & 6-1-2 \end{bmatrix}$$

$$AB_2 \begin{bmatrix} 4 & -1 & -7 & 6 \\ -7 & -5 & 8 & 5 \\ 4 & 0 & 7 & 3 \end{bmatrix}$$

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(c) $A = \begin{bmatrix} 0 & -1 \\ 7 & 2 \\ -4 & -3 \end{bmatrix}$, $B = \begin{bmatrix} 4 & -1 & 2 & 3 & 0 \\ -2 & 0 & 3 & 4 & 1 \end{bmatrix}$

$$AB = \begin{bmatrix} (0\times4)+(-1\times-2) & (0\times-1)+(-1\times0) & (0\times2)+(-1\times3) & (0\times3)+(-1\times4) & (0\times0)+(-1\times1) \\ (7\times4)+(2\times-2) & (7\times-1)+(2\times0) & (7\times2)+(2\times3) & (7\times3)+(2\times4) & (7\times0)+(2\times1) \\ (-4\times4)+(-3\times-2) & (-4\times-1)+(-3\times0) & (-4\times2)+(-3\times3) & (-4\times3)+(-3\times4) & (-4\times0)+(-3\times1) \end{bmatrix}$$

$$AB = \begin{bmatrix} 0+2 & 0+0 & 0-3 & 0-4 & 0-1 \\ 28-4 & -7+0 & 14+6 & 21+8 & 0+2 \\ -16+6 & 4+0 & -8-9 & -12-12 & 0-3 \end{bmatrix}$$

$$AB = \begin{bmatrix} 2 & 0 & -3 & -4 & -1 \\ 24 & -7 & 20 & 29 & 2 \\ -10 & 4 & -17 & -24 & -3 \end{bmatrix}$$

5. Find a matrix A such that

$$\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix} A = \begin{bmatrix} 3 & 0 \\ 1 & 2 \end{bmatrix}$$

Let $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$

$$\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 2a_{11}+3a_{21} & 2a_{12}+3a_{22} \\ a_{11}+4a_{21} & a_{12}+4a_{22} \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 1 & 2 \end{bmatrix}$$

Equating corresponding elements

$2a_{11} + 3a_{21} = 3$ ——— (1)

$2a_{12} + 3a_{22} = 0$ ——— (2)

$a_{11} + 4a_{21} = 1$ ——— (3)

$a_{12} + 4a_{22} = 2$ ——— (4)

By simultaneously solving eq (1), (3) and eq (2), (4)

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Multiplying eq (3) by 2 and subtract from eq (1)

$$2a_{11} + 3a_{21} = 3$$
$$\pm 2a_{11} \pm 8a_{21} = \pm 2$$
$$-5a_{21} = 1$$
$$\boxed{a_{21} = -\frac{1}{5}}$$

putting a_{21} in eq. (3)

$$a_{11} + 4\left(\frac{-1}{5}\right) = 1$$
$$a_{11} - \frac{4}{5} = 1$$
$$a_{11} = 1 + \frac{4}{5}$$
$$\boxed{a_{11} = \frac{9}{5}}$$

Multiplying eq (4) by 2 and subtracting from eq (2)

$$2a_{12} + 3a_{22} = 0$$
$$\pm 2a_{12} \pm 8a_{22} = \pm 4$$
$$-5a_{22} = -4$$
$$\boxed{a_{22} = \frac{4}{5}}$$

Putting a_{22} in eq. (4)

$$a_{12} + 4\left(\frac{4}{5}\right) = 2$$
$$a_{12} + \frac{16}{5} = 2$$
$$a_{12} = 2 - \frac{16}{5}$$
$$\boxed{a_{12} = \frac{-6}{5}}$$

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$$A = \begin{bmatrix} 9/5 & -6/5 \\ -1/5 & +4/5 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}\begin{bmatrix} 9/5 & -6/5 \\ -1/5 & 4/5 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 1 & 2 \end{bmatrix}$$

Find a matrix A such that

$$\begin{bmatrix} 1 & 3 & 2 \\ 2 & 1 & 1 \\ 4 & 0 & 3 \end{bmatrix} A = \begin{bmatrix} 7 & 1 & 3 \\ 1 & 0 & 3 \\ -1 & -3 & 7 \end{bmatrix}$$

Let $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

$$\begin{bmatrix} 1 & 3 & 2 \\ 2 & 1 & 1 \\ 4 & 0 & 3 \end{bmatrix}\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} 7 & 1 & 3 \\ 1 & 0 & 3 \\ -1 & -3 & 7 \end{bmatrix}$$

$$\begin{bmatrix} a_{11}+3a_{21}+2a_{31} & a_{12}+3a_{22}+2a_{32} & a_{13}+3a_{23}+2a_{33} \\ 2a_{11}+a_{21}+a_{31} & 2a_{12}+a_{22}+a_{32} & 2a_{13}+a_{23}+a_{33} \\ 4a_{11}+3a_{31} & 4a_{12}+3a_{32} & 4a_{13}+3a_{33} \end{bmatrix} = \begin{bmatrix} 7 & 1 & 3 \\ 1 & 0 & 3 \\ -1 & -3 & 7 \end{bmatrix}$$

Equating Corresponding elements.

$a_{11}+3a_{21}+2a_{31} = 7$ ——— (1)

$a_{12}+3a_{22}+2a_{32} = 1$ ——— (2)

$a_{13}+3a_{23}+2a_{33} = 3$ ——— (3)

$2a_{11}+a_{21}+a_{31} = 1$ ——— (4)

$2a_{12}+a_{22}+a_{32} = 0$ ——— (5)

$2a_{13}+a_{23}+a_{33} = 3$ ——— (6)

$4a_{11}+3a_{31} = -1$ ——— (7)

$4a_{12}+3a_{32} = -3$ ——— (8)

$4a_{13}+3a_{33} = 7$ ——— (9)

By simultaneously solving eq (1), (4), (7), eq (2), (5), (8) and eq (3), (6), (9)

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$$a_{11} + 3a_{21} + 2a_{31} = 7 \quad \text{— (1)}$$
$$2a_{11} + a_{21} + a_{31} = 1 \quad \text{— (4)}$$
$$4a_{11} + 3a_{31} = -1 \quad \text{— (7)}$$

Multiply eq. (4) by 3 and subtract from eq (1)

$$a_{11} + 3a_{21} + 2a_{31} = 7$$
$$\pm 6a_{11} \pm 3a_{21} \pm 3a_{31} = \pm 3$$
$$-5a_{11} \quad - a_{31} = 4 \quad \text{— (a)}$$

Multiply eq. (a) by 3 and add (a) & (7)

$$-15a_{11} - 3a_{31} = 12$$
$$4a_{11} + 3a_{31} = -1$$
$$-11a_{11} = 11$$
$$a_{11} = \frac{11}{-11}$$
$$\boxed{a_{11} = -1}$$

put a_{11} in eq. (7)

$$4(-1) + 3a_{31} = -1$$
$$-4 + 3a_{31} = -1$$
$$3a_{31} = -1 + 4$$
$$3a_{31} = 3$$
$$\boxed{a_{31} = 1}$$

put a_{11} & a_{31} in eq. (1)

$$-1 + 3a_{21} + 2(1) = 7$$
$$-1 + 3a_{21} + 2 = 7$$
$$3a_{21} + 1 = 7$$
$$3a_{21} = 7 - 1$$
$$3a_{21} = 6$$
$$\boxed{a_{21} = 2}$$

$$a_{12} + 3a_{22} + 2a_{32} = 1 \quad \text{— (2)}$$
$$2a_{12} + a_{22} + a_{32} = 0 \quad \text{— (5)}$$
$$4a_{12} + 3a_{32} = -3 \quad \text{— (8)}$$

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Multiply eq (5) by (3) and subtract from eq(2)

$$a_{12} + 3a_{22} + 2a_{32} = 1$$
$$\pm 6a_{12} \pm 3a_{22} \pm 3a_{32} = 0$$
$$-5a_{12} \quad - a_{32} = 1 \quad \text{—— (b)}$$

Multiply eq. (b) by (3) and add eq (b), (8)

$$-15a_{12} - 3a_{32} = 3$$
$$4a_{12} + 3a_{32} = -3$$
$$-11a_{12} = 0$$
$$\boxed{a_{12} = 0}$$

put a_{12} in eq. (8)

$$4(0) + 3a_{32} = -3$$
$$3a_{32} = -3$$
$$\boxed{a_{32} = -1}$$

put a_{12} and a_{32} in eq. (2)

$$0 + 3a_{22} + 2(-1) = 1$$
$$3a_{22} - 2 = 1$$
$$3a_{22} = 1 + 2 = 3$$
$$\boxed{a_{22} = 1}$$

$$a_{13} + 3a_{23} + 2a_{33} = 3 \quad \text{—— (3)}$$
$$2a_{13} + a_{23} + a_{33} = 3 \quad \text{—— (6)}$$
$$4a_{13} + 3a_{33} = 7 \quad \text{—— (9)}$$

Multiply eq. (6) by 3 and subtract from eq. 3

$$a_{13} + 3a_{23} + 2a_{33} = 3$$
$$\pm 6a_{13} \pm 3a_{23} \pm 3a_{33} = \pm 9$$
$$-5a_{13} \quad - a_{33} = -6 \quad \text{—— (c)}$$

Multiply eq (c) by 3 and add eq. (c), (9)

$$-15a_{13} - 3a_{33} = -18$$
$$4a_{13} + 3a_{33} = 7$$
$$-11a_{13} = -11$$
$$\boxed{a_{13} = 1}$$

put a_{13} in eq (9)

$$4(1) + 3a_{33} = 7$$

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$4 + 3a_{33} = 7$

$3a_{33} = 7 - 4 = 3$

$\boxed{a_{33} = 1}$

put a_{13} and a_{33} in eq. (3)

$1 + 3a_{23} + 2(1) = 3$

$1 + 3a_{23} + 2 = 3$

$3a_{23} + 3 = 3 \neq 1$

$3a_{23} = 3 - 3 = 0$

$\boxed{a_{23} = 0}$

$$A = \begin{bmatrix} -1 & 0 & 1 \\ 2 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & 2 \\ 2 & 1 & 1 \\ 4 & 0 & 3 \end{bmatrix} \begin{bmatrix} -1 & 0 & 1 \\ 2 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 1 & 3 \\ 1 & 0 & 3 \\ -1 & -3 & 7 \end{bmatrix}$$

26. Let

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Find

(a) $A \vee B$

$$A \vee B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \vee \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$A \vee B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

(b) $A \wedge B$

$$A \wedge B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \wedge \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$A \wedge B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

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(c) $A \odot B$

$$A \odot B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \odot \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} (1 \wedge 0) \vee (1 \wedge 1) & (1 \wedge 1) \vee (1 \wedge 0) \\ (0 \wedge 0) \vee (1 \wedge 1) & (0 \wedge 1) \vee (1 \wedge 0) \end{bmatrix}$$

$$= \begin{bmatrix} 0 \vee 1 & 1 \vee 0 \\ 0 \vee 1 & 0 \vee 0 \end{bmatrix}$$

$$A \odot B = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

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